CONSULTATION NOTE: Evaluation of progressive asymmetric weakness, muscle wasting, and muscle twitching in the right upper extremity.

HISTORY OF PRESENT ILLNESS: A 59-year-old male presents with a nine-month history of painless, progressive weakness in his right hand. He initially noted difficulty turning keys and using a screwdriver. Over the past four months, the weakness has spread proximally to involve his right shoulder, and he has noticed prominent muscle twitching (fasciculations) across his chest, back, and arms. In the past six weeks, he has developed occasional slurring of speech when fatigued and choking spells when drinking water. He denies sensory loss, numbness, bowel or bladder incontinence, or cognitive changes.

NEUROLOGICAL ASSESSMENT:

Mental Status: Intact executive functioning, normal language, MoCA score 29/30. Emotional lability noted with inappropriate pseudobulbar laughter during interview.

Cranial Nerves: Mild tongue atrophy with active fasciculations observed along the lateral borders of the tongue at rest. Weakness of the palate and reduced gag reflex. Jaw jerk reflex is hyperactive.

Motor Examination:

- Inspection: Marked atrophy of the right first dorsal interosseous (FDI), thenar eminence, and right deltoid muscles. Widespread spontaneous fasciculations observed in the right biceps, triceps, pectoral muscles, and quadriceps.

- Muscle Strength: Right hand intrinsic muscles 3/5, right wrist extension 3/5, right deltoid 4/5. Left upper extremity 4+/5 throughout. Bilateral lower extremity strength 4+/5.

- Muscle Tone: Spastic catch present in the bilateral lower extremities; normal tone in left arm, flaccid tone in right hand.

Reflexes (Mixed Upper and Lower Motor Neuron Signs):

- Biceps, triceps, and brachioradialis reflexes are brisk (3+) in both upper extremities, despite prominent muscle wasting in the right arm.

- Patellar and Achilles reflexes are pathologically brisk (4+) with sustained bilateral ankle clonus.

- Hoffman sign positive bilaterally; Babinski response is extensor bilaterally.

Sensory: Intact to light touch, pinprick, vibration, and proprioception in all dermatomes.

DIAGNOSTIC TESTING:

Electromyography and Nerve Conduction Studies (EMG/NCS): Normal sensory nerve action potentials (SNAPs). Motor studies show reduced CMAP amplitudes in the right median and ulnar nerves with preserved conduction velocities. Concentric needle EMG demonstrates widespread active denervation (fibrillation potentials and positive sharp waves) and chronic reinnervation (high-amplitude, long-duration, polyphasic motor unit action potentials) in four contiguous body segments: bulbar (tongue), cervical (right FDI, deltoid), thoracic (paraspinal muscles), and lumbosacral (tibialis anterior).

Cervical Spine MRI: Excludes cervical spondylotic myelopathy or structural root compression.

ASSESSMENT:

1. Amyotrophic Lateral Sclerosis (ALS / Motor Neuron Disease), meeting Gold Coast and revised El Escorial criteria for Definite ALS with combined upper and lower motor neuron signs across multiple neuroaxis segments.